ZEN HAMAM

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Personal profile
Junior Gameplay / Systems Programmer with a BSc in Game Software Engineering, specialising in C++ gameplay systems, engine-level programming, and Unreal Engine 5 development. Experienced in building modular game systems and custom engines. Strong foundation in OOP, state machines, and performance aware design, with a focus on clean, maintainable code and playable features. Fluent in English, Italian, and Arabic.

TECHNICAL SKILLS

- **Languages**: C++, C#, Python
- **Frameworks/Engines:** Unity, Unreal Engine 5, SDL2 (engine development)
- **Tools:** Git, GitHub, Debugging, version control, Agile / iterative development
- **Gameplay & Systems**: OOP, State Machines, Multiplayer Replication fundamentals, Data driven design, Input systems, Collision & physics handling

EDUCATION

BSc (Hons) Game Software Engineering — Bournemouth University (2021–2025)
Key modules: AI Systems, Game Programming, Software Design, Physics Simulations

PROJECTS & EXPERIENCE

Code Cadets — Coding Club Instructor — Feb 2026 - Present

- Deliver structured coding lessons in Python, Scratch, HTML, and Minecraft modding to students at private schools across London. Adapt technical content to students of varying ages and experience levels.

Rum Runner's Revenge — C# | Unity | Playable Demo | AI scripts — Jan - May 2024

- Designed and implemented enemy AI using a custom finite state machine (patrol, chase, attack). Built physics based hazards including a collapsing bridge and a live minimap.
- Shipped a playable build in 3 months using Agile sprints and Git version control within a team of 11. Project **nominated for the TIGER Game Award** at Bournemouth University.

2D Arcade Game Engine — C++ | SDL2 | Project Link — Sept 2025 - Ongoing

- Built a modular 2D engine from scratch (rendering, collision, input) shipping two fully playable games: Tetris (with line clearing, row shift logic, next piece preview) and Breakout each with complete game loops and high score tracking.
- Designed reusable OOP architecture with sprite batching for rendering performance and clear separation between engine systems and game specific logic.

Dungeon Slayer — C++ | Unreal Engine 5 | Video Demo — Jun 2025 - Ongoing

- Architected a modular weapon system supporting four combat types (sword, axe, bow, shield) via custom C++ interfaces and a Weapon Component new weapons integrate without touching core character code.
- Integrated GAS with a Player State ASC, separating ability logic from character class.
- Engineered destructible props with physics simulation, custom collision profiles, multi object chain break logic, and timed destruction callbacks via member function pointers.